JEE Mathematics: Quick Review with Grading and Practice Map - Probability (Batch XII)

Grade shows how often the item is tested in the JM exercise of this batch, the JA exercise and the JEE Main Question Bank taken together (144 questions; a question that appears in two of these is counted once): High = 13 or more questions, Medium = 1 to 12 questions, Low = none. Low only means the item is not tested in these questions; it can still be important for understanding. Practice map lists the questions that test the item in the sheet exercises (J-I, J-II, J-M, J-A, C-I, C-II), the DPPs and the Question Bank (QB). 'Main concept' = the question is mainly about this item; 'Also used in' = the question needs it as a secondary step.

Important Definitions

Question 01

What do you mean by sample space?

Model Answer

Sample space S = set of all possible outcomes of a random experiment; each outcome is a sample point; an event is a subset of S.

Ex: a coin is tossed and a die is thrown: $S = \{H1, \ldots, H6, T1, \ldots, T6\}$, $n(S) = 12$.

Grade - Medium

Practice map, main concept: Sheet: J-M 32

Also used in: DPP: 2(11) | Question Bank: 243

Question 02

What do you mean by impossible and sure event?

Model Answer

Impossible event: ϕ, probability 0. Sure event: S itself, probability 1.

Ex: two dice: sum = 13 is impossible ($P = 0$); sum ≤ 12 is sure ($P = 1$).

Grade - Medium

Practice map, main concept: Question Bank: 242

Also used in: none

Question 03

What do you mean by equally likely events, mutually exclusive events and exhaustive events? What is the condition for two or more events to be mutually exclusive and exhaustive?

Model Answer

Equally likely: same chance. Mutually exclusive: $A_i \cap A_j = \phi$ for $i \neq j$. Exhaustive: $A_1 \cup A_2 \cup \cdots \cup A_n = S$.

Mutually exclusive and exhaustive $\Rightarrow P(A_1) + P(A_2) + \cdots + P(A_n) = 1$.

Ex: die: $\{1, 2\}, \{3, 4\}, \{5, 6\}$ are equally likely, mutually exclusive and exhaustive.

Grade - Medium

Practice map, main concept: Sheet: J-II 40, 46; C-I 19 | DPP: 3(2)

Also used in: Sheet: J-I 6; J-II 32, 35; J-M 36 | DPP: 3(13)

Definition of Probability

Question 04

What is the formula for calculation of probability of an event by classical definition of probability?

Model Answer

$$P(A) = \frac{n(A)}{n(S)} = \frac{\text{favourable outcomes}}{\text{total outcomes}}$$

Valid only when all outcomes are equally likely; count $n(A)$ and $n(S)$ with permutations and combinations.

Ex: two numbers from $1, \ldots, 20$, both even: $\dfrac{{}^{10}C_2}{{}^{20}C_2} = \dfrac{45}{190} = \dfrac{9}{38}$.

Grade - High

Practice map, main concept: Sheet: J-I 1, 3, 28; J-II 17; J-M 1-2, 4, 7, 14-15, 24, 26-29, 31, 34, 38, 43; J-A 9-10, 18-20, 34, 38; C-I 1, 5-7, 12, 17, 21; C-II 3-4, 6, 9, 18, 34 | DPP: 1(1-5, 7-9, 11); 2(1, 3-5, 7-9, 12-14); 4(4, 8, 12) | Question Bank: 200, 204, 212, 214-215, 221-222, 224-225, 227, 230, 235, 241, 243-244, 252

Also used in: Sheet: J-I 2, 13, 19; J-II 6, 11; J-M 5, 32; J-A 3, 8, 22-23, 28-29, 36; C-I 9, 19; C-II 15-16, 22, 26-27 | DPP: 1(10, 12); 2(2, 6, 10, 15); 3(11); 4(3, 11); 7(9) | Question Bank: 245, 254

Question 05

What conditions must the probabilities satisfy (bounds on $P(A)$, $P(A \cap B)$, $P(A \cup B)$)?

Model Answer

$0 \le P(A) \le 1$; for mutually exclusive events $\sum P(A_i) \le 1$ (each value also in $[0, 1]$).

$$\max\{0,\ P(A)+P(B)-1\} \le P(A \cap B) \le \min\{P(A), P(B)\} \le \max\{P(A), P(B)\} \le P(A \cup B) \le P(A)+P(B)$$

Ex: $P(A) = 0.6,\ P(B) = 0.8 \Rightarrow 0.4 \le P(A \cap B) \le 0.6$, so $P(A|B) \le \dfrac{0.6}{0.8} = \dfrac{3}{4}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 6; J-M 36; J-A 30 | DPP: 3(5-6)

Also used in: none

Question 06

What is the relation between probabilities of an event and its complement?

Model Answer

$$P(A') = 1 - P(A)$$

Ex: $P(A) = \dfrac{2}{7} \Rightarrow P(A') = \dfrac{5}{7}$.

Grade - Medium

Practice map, main concept: none

Also used in: Sheet: J-I 7, 9 | DPP: 1(6); 3(2); 5(5) | Question Bank: 234

Question 07

When to use complementary method in probability?

Model Answer

When the event has many cases but its opposite has few: "at least one", "not all", "problem is solved", "target is hit". Find $1 - P(\text{opposite})$.

Ex: at least one six in 3 throws: $1 - \left(\frac{5}{6}\right)^3 = \frac{91}{216}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 16; J-II 1, 12, 18, 20; J-A 3, 5, 28-29 | DPP: 1(13); 6(4)

Also used in: Sheet: J-I 27; J-II 16; J-M 9; J-A 11; C-II 34 | DPP: 2(5); 3(12); 5(2, 12); 7(12) | Question Bank: 212

Question 08

How to deal "at least" and "at most" in questions of probability?

Model Answer

At least r: $P(X \geq r) = 1 - P(X \leq r-1)$ (use when it has fewer cases). At most r: $P(X \leq r) = P(0) + P(1) + \cdots + P(r)$.

Ex: two dice, total at least 5: $1 - P(2, 3, 4) = 1 - \frac{1+2+3}{36} = \frac{5}{6}$.

Grade - Medium

Practice map, main concept: none

Also used in: Sheet: J-I 25; J-II 7, 19, 27, 38; J-M 8, 23, 26, 30; J-A 25; C-II 7 | DPP: 7(13) | Question Bank: 204

Question 09

Whether things given are to be considered as alike or distinct?

Model Answer

Always treat objects as distinct (balls of one colour, letters, identical coins) so that the outcomes are equally likely; count numerator and denominator the same way.

Ex: 3 white, 6 red balls, two drawn, both red: $\frac{{}^6C_2}{{}^9C_2} = \frac{15}{36} = \frac{5}{12}$.

Grade - Medium

Practice map, main concept: none

Also used in: Sheet: J-I 15, 28; J-II 4; J-M 1; J-A 18; C-II 2 | DPP: 1(5); 2(4) | Question Bank: 200

Odds

Question 10

How to calculate odds in favour of an event and odds against an event from the probability of the event?

Model Answer

$$\text{Odds in favour} = \frac{P}{1-P} = \frac{\text{favourable}}{\text{unfavourable}}, \qquad \text{odds against} = \frac{1-P}{P}$$

Ex: $P = \frac{2}{7}$: odds in favour $2 : 5$, odds against $5 : 2$.

Grade - Low

Practice map, main concept: Sheet: J-II 2 | DPP: 1(6, 12); 4(6)

Also used in: none

Question 11
How to calculate probability of the event from the odds in favour of an event and odds against an event?

Model Answer
Odds in favour $p : q \Rightarrow P = \frac{p}{p+q}$; odds against $p : q \Rightarrow P = \frac{q}{p+q}$.

Ex: odds against A are $4 : 3 \Rightarrow P(A) = \frac{3}{7}$.

Grade - Low
Practice map, main concept: Sheet: J-II 3 | DPP: 5(6, 13)
Also used in: Sheet: J-II 2; C-II 17 | DPP: 6(14)

Creation of Sample Space

Question 12
Can you create sample space when a die is thrown one/two times?

Model Answer
One die: $\{1, \ldots, 6\}$, 6 outcomes. Two dice (or one die twice): ordered pairs (i, j), $6 \times 6 = 36$ outcomes; n dice: 6^n.

Ex: $P(\text{doublet}) = \frac{6}{36} = \frac{1}{6}$.

Grade - Medium
Practice map, main concept: none
Also used in: Sheet: J-II 1; J-M 34 | DPP: 1(8); 4(12) | Question Bank: 227

Question 13
Can you create sample space when a coin is tossed one/two/three/four times?

Model Answer
n tosses: 2^n sequences of H and T (2, 4, 8, 16); $P(\text{exactly } r \text{ heads}) = \frac{{}^nC_r}{2^n}$.

Ex: 4 tosses, exactly 2 heads: $\frac{{}^4C_2}{16} = \frac{3}{8}$.

Grade - Low
Practice map, main concept: DPP: 3(12)
Also used in: Sheet: J-II 40 | DPP: 6(13)

Question 14
When two dice are thrown do you know the pattern in probabilities when sum of the numbers on them is 2 to 12?

Model Answer
Ways for sum s: $s - 1$ for $2 \le s \le 7$ and $13 - s$ for $7 \le s \le 12$: $1, 2, 3, 4, 5, 6, 5, 4, 3, 2, 1$ (out of 36).

Ex: $P(\text{sum } 9) = \frac{4}{36} = \frac{1}{9}$; sum 7 is the most likely $(\frac{1}{6})$.

Grade - Medium
Practice map, main concept: Sheet: J-I 4 | DPP: 3(1)
Also used in: Sheet: J-I 12, 26; J-A 26; C-II 9 | Question Bank: 235

Venn Diagram

Question 15

Can you draw Venn diagram of 2 events and how to use it in questions?

Model Answer

Four regions: $a = P(A \text{ only})$, $b = P(B \text{ only})$, $c = P(A \cap B)$, $d = P(\text{neither})$, $a+b+c+d = 1$; write every given value in a, b, c, d and solve.

Ex: $P(A) = \frac{3}{8},\ P(B) = \frac{1}{2},\ P(A \cap B) = \frac{1}{4}$: $a = \frac{1}{8},\ b = \frac{1}{4},\ d = \frac{3}{8}$.

Grade - Low

Practice map, main concept: Sheet: J-II 42

Also used in: DPP: 3(9, 14)

Question 16

Can you draw Venn diagram of 3 events and how to use it in questions?

Model Answer

Eight regions (only A, only B, only C, three "exactly two" parts, $A \cap B \cap C$, none); fill from the centre outwards.

$$P(\text{exactly two}) = \sum P(A \cap B) - 3P(A \cap B \cap C), \quad P(\text{at least two}) = \sum P(A \cap B) - 2P(A \cap B \cap C)$$

$$P(\text{exactly one}) = \sum P(A) - 2\sum P(A \cap B) + 3P(A \cap B \cap C)$$

Grade - Medium

Practice map, main concept: Sheet: J-I 33; J-M 3; J-A 33

Also used in: Sheet: J-I 8; C-I 14

Question 17

Write formulae of probabilities of union of two/three/four or more events?

Model Answer

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B \cup C) = \sum P(A) - \sum P(A \cap B) + P(A \cap B \cap C)$$

Ex: $P(A) = 0.25,\ P(B) = 0.5,\ P(A \cap B) = 0.14$: $P(A \cup B) = 0.61$, neither $= 0.39$.

Grade - Medium

Practice map, main concept: Sheet: J-I 9; C-I 20 | DPP: 1(10); 2(11); 3(9, 13-14) | Question Bank: 229

Also used in: Sheet: J-I 17, 33, 39; J-II 3, 31, 36, 45-46; J-A 30, 33; C-II 5 | DPP: 3(5); 6(1, 7)

Question 18

Do you know the pattern in formula of probability of union of events?

Model Answer

$$P(A_1 \cup \cdots \cup A_n) = S_1 - S_2 + S_3 - \cdots + (-1)^{n-1} S_n$$

S_k = sum of probabilities of all intersections of k events (${}^nC_k$ terms); signs alternate.

Ex: 1 to 50, multiple of 4, 6 or 7: $\frac{12+8+7-4-1-1+0}{50} = \frac{21}{50}$.

Grade - Medium

Practice map, main concept: none

Also used in: Sheet: J-A 20 | Question Bank: 229

Question 19
Do you know other ways of expressing union/intersection/difference of two events?

Model Answer
$A \cup B = A + B =$ "A or B" (at least one); $A \cap B = AB =$ "A and B"; $A - B = A \cap B' =$ "only A".

$$P(A \cap B') = P(A) - P(A \cap B), \quad P(A' \cap B') = 1 - P(A \cup B), \quad P(A' \cup B') = 1 - P(A \cap B)$$

Ex: $P(A) = 0.7,\ P(A \cap B') = 0.5 \Rightarrow P(A \cap B) = 0.2.$

Grade - Medium

Practice map, main concept: Sheet: J-I 5; J-II 41 | Question Bank: 234
Also used in: Sheet: J-II 13, 42; J-M 3, 35, 42 | DPP: 3(4); 5(7) | Question Bank: 207, 210

Conditional Probability

Question 20
What is conditional probability?

Model Answer

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0$$

Ex: die, odd number shown; $P(\text{prime}|\text{odd}) = \frac{n\{3,5\}}{n\{1,3,5\}} = \frac{2}{3}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 8; J-A 1; C-I 3; C-II 28 | DPP: 3(3, 8); 5(1) | Question Bank: 207
Also used in: Sheet: J-II 28; J-M 11, 16, 20; J-A 27, 31, 41-42 | DPP: 3(6-7); 6(5) | Question Bank: 253

Question 21
How a conditional probability can be expressed in various ways?

Model Answer

$$P(A \cap B) = P(B)\,P(A|B) = P(A)\,P(B|A), \qquad \frac{P(A|B)}{P(B|A)} = \frac{P(A)}{P(B)}$$

With the Venn regions: $P(A|B) = \frac{c}{b+c}$, $P(B|A') = \frac{b}{b+d}$, $P(A|A \cup B) = \frac{a+c}{a+b+c}$.

Ex: $P(A|B) = \frac{1}{2},\ P(B|A) = \frac{1}{3},\ P(A \cap B) = \frac{1}{6} \Rightarrow P(B) = \frac{1}{3},\ P(A) = \frac{1}{2}$.

Grade - Medium

Practice map, main concept: Sheet: J-M 21, 42; J-A 2, 17; C-I 13 | DPP: 3(10, 15); 6(5) | Question Bank: 210
Also used in: none

Question 22
How to read when a conditional probability expression is given?

Model Answer
$P(A|B)$: "probability of A given that B has occurred"; the event after the bar is the condition (already known), the event before it is asked.

Ex: "11 was picked given that the sum was even" $= P(11 \text{ picked} \,|\, \text{sum even})$.

Grade - Medium

Practice map, main concept: Sheet: J-A 42
Also used in: none

Question 23
What happens to the sample space in case of conditional probability questions?

Model Answer

The given event B becomes the new (reduced) sample space: $P(A|B) = \dfrac{n(A \cap B)}{n(B)}$ for equally likely outcomes.

Ex: two dice, total 8: $(2,6), (3,5), (4,4), (5,3), (6,2)$; $P(\text{second die } 5) = \dfrac{1}{5}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 2, 10-13; J-II 25, 43, 49; J-A 22, 26, 36; C-II 29 | DPP: 3(7, 11)
Also used in: Sheet: J-II 5

Question 24
What to keep in mind while solving conditional probability questions?

Model Answer

Identify which event is given. $P(A'|B) = 1 - P(A|B)$, but $P(A|B') \neq 1 - P(A|B)$ in general; $P(A|B) \neq P(B|A)$.

Ex: $P(A) = \dfrac{1}{2}$, $P(B) = \dfrac{1}{3}$, $P(A \cap B) = \dfrac{1}{4}$: $P(A'|B) = \dfrac{1}{4}$, $P(A|B') = \dfrac{1/4}{2/3} = \dfrac{3}{8}$.

Grade - Medium

Practice map, main concept: Sheet: J-M 35; C-I 18
Also used in: Sheet: J-II 47; J-A 17; C-I 13; C-II 25 | DPP: 7(15)

Question 25
What is conditional probability when A is subset of B?

Model Answer

$$A \subseteq B: \quad P(A|B) = \frac{P(A)}{P(B)}, \qquad P(B|A) = 1$$

Ex: die, $A = \{2\}$, $B = \text{even}$: $P(A|B) = \dfrac{1}{3}$, $P(B|A) = 1$.

Grade - Low

Practice map, main concept: none
Also used in: DPP: 3(8)

Question 26
What is conditional probability when A and B are disjoint?

Model Answer

$$A \cap B = \phi: \quad P(A|B) = P(B|A) = 0$$

Ex: die: $P(\text{odd}|\text{even}) = 0$.

Grade - Low

Practice map, main concept: none
Also used in: none

Multiplication Theorem

Question 27

How to use multiplication theorem and what to keep in mind?

Model Answer

$$P(A \cap B) = P(A)\,P(B|A), \qquad P(A_1 \cap A_2 \cap A_3) = P(A_1)\,P(A_2|A_1)\,P(A_3|A_1 \cap A_2)$$

Draws one by one: without replacement the bag changes after every draw; with replacement it does not. Multiply along the order, add over different orders.

Ex: 3 white, 6 red, without replacement, first white then red: $\frac{3}{9} \cdot \frac{6}{8} = \frac{1}{4}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 14-15; J-II 4-5, 8; J-M 12, 22; J-A 15-16, 32, 41; C-II 2, 16-17, 22 | DPP: 2(6); 3(4); 4(1); 5(3) | Question Bank: 211, 228

Also used in: Sheet: J-A 6, 24; C-I 8; C-II 32 | DPP: 4(2, 6, 9); 5(10); 6(12, 15) | Question Bank: 208, 252

Question 28

When to use ${}^{n}C_{r}$ method?

Model Answer

When balls are drawn without replacement and the order is not asked (drawn together or "one is ..., one is ..."): $P = \frac{\text{ways to select favourable}}{{}^{n}C_{r}}$; it equals the one-by-one answer.

Ex: 3 white, 6 red, two drawn, one of each: $\frac{{}^{3}C_{1}\,{}^{6}C_{1}}{{}^{9}C_{2}} = \frac{1}{2} = 2! \cdot \frac{3}{9} \cdot \frac{6}{8}$.

Grade - Medium

Practice map, main concept: Sheet: J-II 6-7; C-II 26-27 | DPP: 2(2, 10, 15); 4(2-3, 11)

Also used in: Sheet: J-II 4, 17-18, 21; J-A 7; C-I 17; C-II 3, 6, 10 | DPP: 1(11, 13); 7(8) | Question Bank: 228

Independent Events

Question 29

What do you mean when 2 or more events are independent?

Model Answer

Occurrence of one does not change the chance of the other: $P(A|B) = P(A)$. Separate tosses, throws, shots, draws with replacement are independent.

Independent $\neq$ mutually exclusive: mutually exclusive events with non-zero probabilities are never independent.

Grade - High

Practice map, main concept: Sheet: J-II 10-11, 31, 34; J-A 6, 8; C-II 1, 15, 32 | DPP: 6(2, 7)

Also used in: Sheet: J-I 14, 16; J-II 8, 12, 47; J-M 12-13, 17; J-A 1, 4-5, 9, 15-16, 32; C-I 12; C-II 8 | DPP: 4(4, 8); 5(1, 3, 6, 9); 6(4) | Question Bank: 211

Question 30
How to check whether 2 events are independent?

Model Answer

$$A, B \text{ independent} \iff P(A \cap B) = P(A)\,P(B)$$

Then A, B'; A', B; A', B' are also independent.

Ex: card: King and diamond: $\frac{1}{52} = \frac{1}{13} \cdot \frac{1}{4}$, independent.

Grade - Medium

Practice map, main concept: Sheet: J-I 7, 17, 19; J-II 35-36, 45; J-A 23; C-I 16 | DPP: 5(7); 6(1, 10-11, 13)

Also used in: Sheet: J-I 8; J-II 39, 44, 48; J-M 10, 21; J-A 2; C-I 20 | DPP: 3(15)

Question 31
How to check whether 3 events are independent?

Model Answer

All four: $P(A \cap B) = P(A)P(B)$, $P(B \cap C) = P(B)P(C)$, $P(C \cap A) = P(C)P(A)$ and $P(A \cap B \cap C) = P(A)P(B)P(C)$. The first three alone = pairwise independent.

Ex: two coins, A: head on 1st, B: head on 2nd, C: exactly one head: pairwise independent, but $P(A \cap B \cap C) = 0 \neq \frac{1}{8}$.

Grade - Low

Practice map, main concept: Sheet: J-I 18; J-II 32, 44, 47

Also used in: Sheet: C-I 16

Question 32
How to use Venn diagram in case 2 events are independent?

Model Answer

With $P(A) = p_1$, $P(B) = p_2$: both p_1p_2, only A $p_1(1-p_2)$, only B $(1-p_1)p_2$, neither $(1-p_1)(1-p_2)$; exactly one $= p_1 + p_2 - 2p_1p_2$.

Ex: $p_1 = \frac{1}{2}$, $p_2 = \frac{2}{3}$: problem solved $= 1 - \frac{1}{2} \cdot \frac{1}{3} = \frac{5}{6}$.

Grade - Medium

Practice map, main concept: Sheet: J-II 13, 39; J-M 10 | DPP: 5(5, 11)

Also used in: none

Question 33
How to use Venn diagram in case 3 events are independent?

Model Answer

Each region is a product: only $A = p_1q_2q_3$, $A \cap B$ only $= p_1p_2q_3$, all $= p_1p_2p_3$, none $= q_1q_2q_3$ $(q_i = 1 - p_i)$.

Ex: $\frac{1}{6}, \frac{1}{4}, \frac{1}{3}$: at least one $= 1 - \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{2}{3} = \frac{7}{12}$.

Grade - Medium

Practice map, main concept: Sheet: J-A 4; C-I 14; C-II 5 | DPP: 5(9)

Also used in: DPP: 5(13)

Question 34
How to solve infinite GP pattern based questions in independent events?

Model Answer
Trials repeated until a success (players in turn): first success at trial k: $q^{k-1}p$; $P(\text{more than } k \text{ trials}) = q^k$.

$$P(\text{first of two players wins}) = p + q^2p + q^4p + \cdots = \frac{p}{1-q^2}$$

Ex: Ram and Shyam throw a die in turn, face 1 wins, Ram first: $\dfrac{1/6}{1-25/36} = \dfrac{6}{11}$.

Grade - Medium
Practice map, main concept: Sheet: J-I 35; J-II 9, 26-28; J-M 11; J-A 35; C-I 10 | DPP: 5(4, 8); 6(12) | Question Bank: 213, 253-254
Also used in: Sheet: J-II 48; J-M 6 | Question Bank: 202, 255

Law of Total Probability

Question 35
How to solve questions of law of total probability?

Model Answer

$$P(A) = \sum_{i=1}^{n} P(B_i)\,P(A|B_i), \quad B_1, \ldots, B_n \text{ mutually exclusive and exhaustive}$$

Ex: 3 fair coins and 1 double-headed coin, one drawn and tossed: $P(H) = \dfrac{3}{4}\cdot\dfrac{1}{2} + \dfrac{1}{4}\cdot 1 = \dfrac{5}{8}$.

Grade - Medium
Practice map, main concept: Sheet: J-I 20-22, 32; J-II 14, 16, 21, 23, 29; J-A 13, 31; C-I 8, 11; C-II 8, 20, 23, 31 | DPP: 4(7, 9-10); 6(8, 14-15); 7(7-10) | Question Bank: 218, 240
Also used in: Sheet: J-II 25, 43, 49; J-M 44; J-A 21, 39, 43; C-I 18; C-II 1, 14, 19 | DPP: 4(1)

Question 36
Can you solve law of total probability questions by drawing tree diagram?

Model Answer
First branches: the cases B_i with $P(B_i)$; second branches: $P(A|B_i)$. Multiply along each path ending in A and add the paths.

Ex: coins above: paths fair$\to H$: $\dfrac{3}{8}$, double-headed$\to H$: $\dfrac{1}{4}$; total $\dfrac{5}{8}$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 37
How to express solution of questions of law of total probability in board method?

Model Answer
Write: let $E_1, E_2, \ldots$ be the cases (mutually exclusive, exhaustive) and A the event; state $P(E_i)$ and $P(A|E_i)$; then $P(A) = \sum P(E_i)P(A|E_i)$.

Ex: purse compartments $1R2P$, $2R3P$, $3R4P$: $P(R) = \dfrac{1}{3}\left(\dfrac{1}{3} + \dfrac{2}{5} + \dfrac{3}{7}\right) = \dfrac{122}{315}$.

Grade - Low
Practice map, main concept: none

Also used in: none

Bayes' Theorem

Question 38

How to identify whether a question is of Bayes' theorem?

Model Answer

The result (effect) is given and the probability of a cause is asked: "given that the ball is red, find the probability that it came from bag II".

$$P(B_i|A) = \frac{P(B_i)\,P(A|B_i)}{\sum_j P(B_j)\,P(A|B_j)}$$

Ex: compartments above, coin is a rupee: $P(\text{3rd}|R) = \dfrac{\frac{1}{3}\cdot\frac{3}{7}}{122/315} = \dfrac{45}{122}$.

Grade - High

Practice map, main concept: Sheet: J-I 23, 29-31; J-II 15, 22, 24, 30; J-M 17-18, 41, 44; J-A 7, 12, 14, 21, 39, 43; C-II 10-12, 14, 19, 24-25, 30, 33 | DPP: 7(5-6, 14-17) | Question Bank: 201, 205, 208, 217, 223, 226, 246

Also used in: Sheet: J-II 49; C-II 13, 20, 36 | DPP: 7(4) | Question Bank: 233

Question 39

How to write denominator in Bayes' theorem questions?

Model Answer

Denominator = total probability of the observed event: all paths that lead to the observation. Numerator = only the path of the asked cause.

Ex: 4 fair coins, 1 double-headed; HH seen: $P(\text{fair}|HH) = \dfrac{\frac{4}{5}\cdot\frac{1}{4}}{\frac{4}{5}\cdot\frac{1}{4}+\frac{1}{5}\cdot 1} = \dfrac{1}{2}$.

Grade - Medium

Practice map, main concept: none

Also used in: Sheet: J-I 23, 29, 31; J-II 15, 24, 30; J-M 18; J-A 12; C-II 11-12, 24, 30, 33 | DPP: 7(14, 16-17) | Question Bank: 205

Question 40

How to use extended Bayes' theorem and when to use?

Model Answer

When, after an observation A, the probability of a further event C (next toss/draw) is asked: $P(C|A) = \sum_i P(B_i|A)\,P(C|B_i \cap A) = \dfrac{\sum P(B_i)P(A\cap C|B_i)}{\sum P(B_i)P(A|B_i)}$.

Ex: 3 fair and 2 double-headed coins; head seen, same coin tossed again: $\dfrac{\frac{3}{5}\cdot\frac{1}{4}+\frac{2}{5}}{\frac{3}{5}\cdot\frac{1}{2}+\frac{2}{5}} = \dfrac{11}{14}$.

Grade - Medium

Practice map, main concept: Sheet: C-II 36 | DPP: 7(4) | Question Bank: 233

Also used in: Sheet: C-II 31 | Question Bank: 246

Binomial Probability Distribution (BPD)

Question 41
What are the conditions for a trial to be Bernoulli trials?

Model Answer
(i) finite number n of trials, (ii) trials independent, (iii) each trial has two outcomes, success or failure, (iv) $P(\text{success}) = p$ same in every trial.

Ex: draws with replacement are Bernoulli trials; draws without replacement are not.

Grade - Low
Practice map, main concept: DPP: 6(3)
Also used in: DPP: 6(2)

Question 42
What is X and q in BPD?

Model Answer
X = number of successes in n trials ($X = 0, 1, \ldots, n$); $p = P(\text{success})$, $q = 1 - p = P(\text{failure})$.
Ex: success = "six" on a die: $p = \dfrac{1}{6},\ q = \dfrac{5}{6}$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 43
What is the formula for $P(X = r)$?

Model Answer

$$P(X = r) = {}^{n}C_r\, p^r\, q^{n-r}, \quad r = 0, 1, \ldots, n$$

Ex: a die thrown 5 times, 3 appears exactly twice: ${}^{5}C_2 \left(\dfrac{1}{6}\right)^2 \left(\dfrac{5}{6}\right)^3 = \dfrac{625}{3888}$.

Grade - High
Practice map, main concept: Sheet: J-I 24-27; J-II 19, 33, 38; J-M 5, 8, 13, 19, 23, 30; J-A 24; C-II 7 | DPP: 5(10); 6(9); 7(1-3, 12-13) | Question Bank: 232, 236, 247
Also used in: Sheet: J-I 38, 40; J-II 20, 37; J-M 25, 33, 37, 39; J-A 11, 25; C-I 11; C-II 35 | DPP: 6(3); 7(5) | Question Bank: 239

Question 44
How is BPD denoted ($B(n, p)$)?

Model Answer
$X \sim B(n, p)$: n trials, success probability p; the probabilities are the terms of $(q + p)^n$, so they add to 1.
Ex: $B(5, \frac{1}{2})$: $P(X = 0) = \dfrac{1}{32}$; $\dfrac{P(X = r)}{P(X = r - 1)} = \dfrac{n - r + 1}{r} \cdot \dfrac{p}{q}$.

Grade - Medium
Practice map, main concept: none
Also used in: Sheet: J-M 19 | DPP: 7(3)

Question 45

How to find the least number of trials n so that the probability of at least one (or at least r) success is at least k?

Model Answer

At least one: $1 - q^n \geq k \iff q^n \leq 1 - k$; take logs or try $n = 1, 2, \ldots$. At least r: $1 - \sum_{i=0}^{r-1} {}^nC_i p^i q^{n-i} \geq k$, test values of n.

Ex: fair coin, $P(\text{at least one head}) \geq 0.9$: $\left(\frac{1}{2}\right)^n \leq 0.1 \Rightarrow n = 4$.

Grade - Medium

Practice map, main concept: Sheet: J-M 9; J-A 11, 25; C-II 35 | DPP: 5(2, 12)

Also used in: none

Question 46

How to find the most probable number of successes in $B(n, p)$?

Model Answer

The most probable r satisfies $(n+1)p - 1 \leq r \leq (n+1)p$; if $(n+1)p$ is an integer, both $(n+1)p - 1$ and $(n+1)p$ are most probable.

Ex: fair coin tossed 99 times: $(n+1)p = 50$, so $r = 49$ and 50.

Grade - Low

Practice map, main concept: Sheet: J-I 36; J-II 37

Also used in: none

Probability through Tree Diagram

Question 47

How to solve questions of probability through tree diagram?

Model Answer

One level per stage; write on each branch its conditional probability (the bag after the earlier stages); required probability = sum of the products along the favourable paths.

Ex: bag 1R, 2B; ball replaced with one more of its colour; three reds: $\frac{1}{3} \cdot \frac{2}{4} \cdot \frac{3}{5} = \frac{1}{10}$.

Grade - Medium

Practice map, main concept: Sheet: J-A 27; C-II 13, 21 | DPP: 4(5); 7(11)

Also used in: Sheet: J-I 23, 32; J-II 29; J-A 13, 35; C-I 10 | DPP: 4(7, 10); 5(4); 7(7, 10) | Question Bank: 218

Question 48

How to compute probability of any branch of a tree?

Model Answer

Probability of a path = product of the probabilities on its branches from the start; the branches leaving one node add up to 1.

Ex: same bag, blue then blue: $\frac{2}{3} \cdot \frac{3}{4} = \frac{1}{2}$.

Grade - Low

Practice map, main concept: none

Also used in: Sheet: J-I 21; C-II 13, 21 | DPP: 4(5); 7(11)

Geometrical Probability

Question 49

How to solve geometrical probability questions?

Model Answer

$$P = \frac{\text{favourable length / area / volume}}{\text{total length / area / volume}}$$

Two real numbers x, y from $[0, a]$: sample space = square of side a; draw the region of the condition.

Ex: point in a circle of radius a, nearer the circumference than the centre: $1 - \dfrac{\pi(a/2)^2}{\pi a^2} = \dfrac{3}{4}$.

Grade - Low

Practice map, main concept: Sheet: C-I 4

Also used in: none

Probability Distribution of Random Variable

Question 50

How to write probability distribution of a random variable for an experiment?

Model Answer

List all values x_i of X and $p_i = P(X = x_i)$ with $p_i \geq 0$, $\sum p_i = 1$ (use this to find an unknown k).

Drawing without replacement: $P(X = r) = \dfrac{{}^{a}C_r \, {}^{b}C_{m-r}}{{}^{a+b}C_m}$.

Ex: two coins, X = number of heads: $0, 1, 2$ with $\dfrac{1}{4}, \dfrac{1}{2}, \dfrac{1}{4}$.

Grade - High

Practice map, main concept: Sheet: J-I 39; J-M 16, 20; J-A 40 | Question Bank: 202, 245

Also used in: Sheet: J-M 6, 22, 40, 45; J-A 37; C-I 3 | Question Bank: 203, 206, 209, 216, 219-220, 231, 237-238, 248-251

Mean and Variance of Probability Distribution

Question 51

How to calculate mean and variance of probability distribution?

Model Answer

$$\mu = E(X) = \sum p_i x_i, \qquad \sigma^2 = \sum p_i x_i^2 - \mu^2, \qquad \sigma = \sqrt{\sigma^2}$$

Expectation of a game = $\sum p_i M_i$ (negative: unfavourable). $E(aX+b) = a\mu+b$, $\mathrm{Var}(aX+b) = a^2\sigma^2$.

Ex: two coins, X = heads: $\mu = 1$, $\sigma^2 = \dfrac{1}{2} + 1 - 1 = \dfrac{1}{2}$.

Grade - High

Practice map, main concept: Sheet: J-I 34; J-M 6, 45; J-A 37 | Question Bank: 203, 206, 209, 216, 219-220, 231, 237-238, 248-251, 255

Also used in: Sheet: J-I 35; J-A 40

Mean and Variance of Binomial Probability Distribution

Question 52

How to calculate mean and variance of binomial probability distribution?

Model Answer

$$\text{Mean} = np, \qquad \text{Variance} = npq, \qquad \text{SD} = \sqrt{npq}$$

Variance $<$ mean; $\dfrac{\text{variance}}{\text{mean}} = q$ gives q, then p and n.

Ex: mean 4, variance 2: $q = \dfrac{1}{2},\ p = \dfrac{1}{2},\ n = 8.$

Grade - Medium

Practice map, main concept: Sheet: J-I 37-38, 40-41; J-M 25, 33, 37, 39-40 | Question Bank: 239

Also used in: Sheet: J-I 36

Miscellaneous

Question 53

How to solve probability questions of units digit in product of n numbers?

Model Answer

Only the last digits matter; each last digit 0 to 9 has probability $\dfrac{1}{10}$. Units digit of the product of n numbers:

$$1, 3, 7, 9: \ \frac{4^n}{10^n}; \quad 2, 4, 6, 8: \ \frac{8^n - 4^n}{10^n}; \quad 5: \ \frac{5^n - 4^n}{10^n}; \quad 0: \ \frac{10^n - 8^n - 5^n + 4^n}{10^n}$$

Ex: two numbers multiplied, units digit 5: $\dfrac{25 - 16}{100} = \dfrac{9}{100}$.

Grade - Low

Practice map, main concept: Sheet: J-II 48; C-I 2, 9, 15 | DPP: 6(6)

Also used in: none